Zeyad Manaa

Graduate Research and Teaching Assistant

Department of Aerospace Engineering
King Fahd University for Petroleum and Minerals
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Research Interests

- Machine learning and data science for modeling and control of dynamical systems
- Control theory, optimization, and AI for dynamical systems
- Model reduction, and feedback control of dynamical systems

Education

2023–present **M.Sc in Aerospace Engineering**, *King Fahd University for Petroleum & Minerals*, Dhahran, Saudi Arabia.
Koopman operator theory, physics informed machine learning, sparse identification of nonlinear dynamics, dynamic mode decomposition, data-driven modelling and control

2017–2022 **B.Sc in Aerospace Engineering**, *University of Science and Technology at Zewail City*, Giza, Egypt.

Thesis Development of the software package for the attitude determination and control algorithm of a cube satellite

Patents and Publications

Under Review and In Preparation Publications

2024 **Zeyad M Manaa**, Syed Saad Azhar Ali, Mohammed Abido, and Ayman M. Abdallah. Local koopman observables for identifying globally linear model for quadrotor's dynamics. *In Prep*, 2024.

2024 **Zeyad M Manaa**, Syed Saad Azhar Ali, Mohammed Abido, and Ayman M. Abdallah. Identification of a globally linear model for quadrotor's dynamics using data-driven koopman theory. *In Prep*, 2024.

2024 **Zeyad M Manaa** and Naef Qassem. Novel airfoil for improved supersonic performance. *In Review in Aerospace Science and Technology*. Elsevier, 2024.

2024 **Zeyad M Manaa** and Ayman M. Abdallah. Robust data-driven identification of quadrotor's equation of motion. *In Prep*, 2024.

In Conference Proceedings

2024 **Zeyad M Manaa**, Mohammed R Elbalshy, and Ayman M Abdallah. Data-driven discovery of the quadrotor equations of motion via sparse identification of nonlinear dynamics. In *AIAA SCITECH 2024 Forum*. AIAA, 2024.

Patents

2024 **Zeyad M Manaa** and Naef Qassem. Efficient airfoil for improved supersonic performance for fighters. *Under Review*, 2024.

Research Experience

January 2023 - Present **Research Assistant - Flight Dynamics and Control Lab**.
King Fahd University for Petroleum and Minerals, Saudi Arabia

Advisor Ayman Abdalla, *Chairman and Assistant Professor, Department of Aerospace Engineering, KFUPM*

- Research on Koopman operator modeling to globally linearize nonlinear dynamics and its intersection with deep learning
- Worked on the basis of reduced order modeling for high dimensional dynamical systems using proper orthogonal decomposition and dynamic mode decomposition.

June 2022 - August 2022 **Research Intern - Aerial Robotics Lab**.
National University of Science and Technology, Pakistan

Advisors Muhammad Jawad Khan and Muhammad Aleem Siddiqui

- Used hand gestures to control drones via built-in/web camera using **Computer Vision/Machine Learning** on experimental setup.
- Developed an autonomous control system for a quad-copter drone for Tello drone.

August 2021 - July 2022 **Bachelor's Thesis Intern - ADCS Lab**.
Egyptian Space Agency – University of Science and Technology at Zewail City, Egypt

Advisor Mustafa Abdalla, *Former Chairman, Department of Aerospace Engineering, Zewail City*

- Developed the software of the **attitude determination and control** algorithm of a cube satellite.
- Dealt with the space environment as a means of Earth's Magnetic Field (IGRF Model), Earth's gravitational Field (Spherical Harmonics) as well as modeling the space disturbances using **MATLAB**.
- Developed an efficient control scheme based on magnetic actuators in terms of maneuver time, robustness, stability, and applicability for implementation on an on-board computer.

Presentations & Talks

March 2024 **Upcoming talk: Data-driven techniques for modeling and control in aerospace applications**, *Aerospace Dept. KFUPM.*

December 2023 **Presentation: Observations are all you need! - Koopman's theory unfolds globally linear model for quadrotor**, *Aerospace Dept. KFUPM.*

May 2023 **Presentation: On proper orthogonal decomposition and dynamic mode decomposition for aerodynamic applications**, Math. Dept. KFUPM.

May 2023 **Presentation: Sparse identification of nonlinear dynamics for quadrotor's equation of motion discovery**, *Aerospace Dept. KFUPM.*

March 2023 **Presentation: Convex optimization for supersonic airfoil design using linear flow theory**, *Aerospace Dept. KFUPM.*

Teaching

Spring, 2024 **AE 540 - Advanced Flight Mechanics**, *Graduate Course*, Aerospace Dept., KFUPM, **Teaching Assistant**.

Fall, 2023 **AE 426 - Introduction to Flight Mechanics**, *Undergrad. Course*, Aerospace Dept., KFUPM, **Instructor**.

Fall, 2023 **AE 315 - Systems and Control**, *Undergrad. Lab*, Aerospace Dept., KFUPM, **Instructor**, (Link).

Fellowships & Awards

2023 ***Data-driven Modeling and Control Research Grant***
King Fahd University for Petroleum and Minerals, Interdisciplinary Research Center for Aviation & Space Exploration.

2023 – present ***Graduate International Research Assistance Scholarship***
King Fahd University for Petroleum and Minerals, Aerospace Engineering department.

2023 Receipt of **Mohammad Al-Aqeel Grant for Graduate Students Conference Attendance** to attend ***AIAA, SCITECH*** conference in Orlando, USA, 2024
King Fahd University for Petroleum and Minerals, Deanship of Research Oversight and Coordination.

2022 Receipt of the **Fully Funded Internship Program for International students** award
National University of Science and Technology.

2021 **Future Work is Digital Scholarship**
Ministry of Communications and Information Technology, Egypt

2017 **Full Undergraduate Scholarship**
University of Science and Technology at Zewail City, Egypt

Academic Achievements & Recognitions

2023 **First Honour Award**
KFUPM first honour for graduate students award for scoring GPA of 4.0/4.0

2019 **Smart City Hackathon**
Global finalist, awarded the **1st place** award.

Skills

Programming Languages	Python, MATLAB, C++, TeX; also basic ability with Julia
Software Skills	Pytorch, tensorflow, OpenCV, Sci-Kit, ROS, and Gazebo
Platforms	Linux, Windows, and MacOS
Relevant Coursework	Machine Learning, Advanced Dynamics, Linear Algebra, Discrete Math, Probability and Statistics, ODE, PDE, Modern Control, Intelligent Control, Algorithms
Professional Memberships	AIAA student member
Other Skills	GitHub, Mathematica, bash-scripting, and LabVIEW
Languages	Arabic (Mother Tongue), English (IELTS: 7.5).

Professional Service

Reviewer for Conference Proceedings

2023 **AIAA, SCITECH Conference**.

Leadership

2020 **Euroavia Egypt**.

Led 10 people as a the Media committee head for the Euroavia Egypt, student branch

2019 **Zewail City Science Festival**.

Led 15 people as a the Media committee head for the Zewail City Science Festival mega even, Egypt.

Referees

Available upon request